

Lognormal Prob Distribution

Defⁿ: Let X be a continuous r.v. taking only positive values.

If $Y = \log_e X$ follows a $N(\mu, \sigma^2)$, then we say that X has a lognormal distribution with parameters μ and σ^2 .

We write $X \sim LN(\mu, \sigma^2)$. The density function of X is given by

$$f_X(x) = \frac{1}{x \sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{\log_e x - \mu}{\sigma} \right)^2}; \quad x > 0$$

where $\mu \in (-\infty, \infty)$ and $\sigma > 0$.

Ex: In many practical situation, it is more realistic. For instance, the weight of manufactured items could have a lognormal distribution.

Ex: $X \sim LN(5, 4)$, Find $P(X \leq 100)$

$$P(X \leq 100) = P(\log_e X \leq \log_e 100) = P(Y \leq \log_e 100)$$

$$Y \sim N(5, 4)$$

$$\Rightarrow P\left(\frac{Y - \mu}{\sigma} \leq \frac{\log_e 100 - 5}{2}\right)$$

$$= P\left(Z \leq \frac{4.605 - 5}{2}\right) = P(Z \leq -0.20)$$

$$= 0.4207 \quad (\text{using normal table})$$



Derivation of p.d.f., Mean & variance of lognormal distⁿ

$$Y = \log_e X \sim N(\mu, \sigma^2)$$

$$f_Y(y) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \frac{(y - \mu)^2}{\sigma^2}}, \quad -\infty < y < \infty$$

$$f_X(x) = f_Y(y) |J| \quad Y = \log_e X \quad X = e^Y, \quad \text{if } Y = -\infty, X = 0$$

$$Y = \infty, X = \infty$$

$$J = \frac{dy}{dx} = \frac{1}{x}$$

$$f_X(x) = \frac{1}{x \cdot \sigma \sqrt{2\pi}} e^{-\frac{1}{2} \frac{(\log_e x - \mu)^2}{\sigma^2}}, \quad x > 0$$

We know the r th moment about origin is given as

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$$\mu_r' = E(X^r) = E(e^{rY})$$

$$= M_Y(r)$$

$$= e^{kr + \frac{1}{2}r^2\sigma^2}$$

$$Y = \log_e X$$

$$\Rightarrow X = e^Y$$

$$Y \sim N(k, \sigma^2)$$

$$\text{Mean } \mu_1' = \frac{e^{k + \frac{\sigma^2}{2}}}{e^{2k + 2\sigma^2}} = e^k \cdot e^{\frac{\sigma^2}{2}}$$

$$\mu_2' = e^{2k + 2\sigma^2}$$

$$\text{Variance } \mu_2 = \mu_2' - \mu_1'^2$$

$$= e^{2k + 2\sigma^2} - e^{2k + \sigma^2}$$

$$= e^{2k} \cdot e^{2\sigma^2} - e^{2k} \cdot e^{\sigma^2}$$

$$= e^{2k} e^{\sigma^2} (e^{\sigma^2} - 1)$$

Hence,

$$\begin{aligned} \text{Mean} &= e^k \cdot e^{\frac{\sigma^2}{2}} \\ \text{Variance} &= e^{2k} \cdot e^{\sigma^2} (e^{\sigma^2} - 1) \end{aligned}$$

Note: Lognormal distⁿ arises in problems of economics, biology, geology and reliability theory etc.

If X and Y are two normally correlated variables with correlation coefficient ρ and means and variances (μ_1, μ_2) and (σ_1^2, σ_2^2) respectively. Then the prob. distⁿ of bivariate normal distⁿ is given as

$$f(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-\left[\frac{1}{2(1-\rho^2)}\left\{\frac{(x-\mu_1)^2}{\sigma_1^2} + \frac{(y-\mu_2)^2}{\sigma_2^2} - 2\rho\frac{(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2}\right\}\right]}$$

$$\begin{aligned} &-\infty < x < \infty, -\infty < y < \infty \\ &-\infty < \mu_1 < \infty, -\infty < \mu_2 < \infty \\ &\sigma_1^2 > 0, \sigma_2^2 > 0 \end{aligned}$$

$\mu_1, \mu_2, \sigma_1^2, \sigma_2^2$ and ρ are the parameters of bivariate normal distⁿ

Remark: $(X, Y) \sim^{BV} N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$. If in particular

$$\mu_1 = \mu_2 = 0, \sigma_1 = \sigma_2 = 1, \text{ then } (X, Y) \sim N(0, 0, 1, 1, \rho) \sim BVN(0, 0, 1, 1, \rho)$$

MGF: $M_{X,Y}(t_1, t_2) = E[e^{t_1X + t_2Y}] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{(t_1x + t_2y)} f(x, y) dx dy$

$$= \exp\left[t_1\mu_1 + t_2\mu_2 + \frac{1}{2}(t_1^2\sigma_1^2 + t_2^2\sigma_2^2 + 2\rho t_1t_2\sigma_1\sigma_2)\right]$$

In particular if $(X, Y) \sim BVN(0, 0, 1, 1, \rho)$, then

$$M_{X,Y}(t_1, t_2) = \exp\left[\frac{1}{2}(t_1^2 + t_2^2 + 2\rho t_1t_2)\right]$$

Ex. If $f(x) = \frac{1}{\theta} e^{-x/\theta} ; x > 0$

$$M_X(t) = E e^{tx} = \int_0^{\infty} e^{tx} \frac{1}{\theta} e^{-x/\theta} dx$$

$$= \frac{1}{\theta} \int_0^{\infty} e^{-(\frac{1}{\theta} - t)x} dx$$

$$= \frac{1}{\theta} \int_0^{\infty} e^{-\frac{(1-\theta t)}{\theta} x} dx$$

$$= \frac{1}{\theta} \left[-e^{-\frac{(1-\theta t)}{\theta} x} \cdot \frac{\theta}{(1-\theta t)} \right]_0^{\infty}$$

$$= \frac{1}{\theta} \cdot \frac{\theta}{(1-\theta t)} = \frac{1}{(1-\theta t)} = (1-\theta t)^{-1}$$

$$M_X(t) = (1-\theta t)^{-1}$$

Mean $h_1' = \frac{\partial M_X(t)}{\partial t} \Big|_{t=0} ; \frac{\partial M_X(t)}{\partial t} = (-1)(1-\theta t)^{-2}(-\theta)$

$$h_1' = \frac{\partial M_X(t)}{\partial t} \Big|_{t=0} = \theta$$

$$h_2' = \frac{\partial^2 M_X(t)}{\partial t^2} \Big|_{t=0}$$

Hence, $\frac{\partial^2 M_X(t)}{\partial t^2} = (-1)(-2)(1-\theta t)^{-3}(-\theta)(-\theta)$

$$h_2' = \frac{\partial^2 M_X(t)}{\partial t^2} \Big|_{t=0} = 2\theta^2$$

$\therefore$ Variance $h_2 = h_2' - h_1'^2 = 2\theta^2 - \theta^2 = \theta^2$

Mean = θ , Variance = θ^2
